

BACKEND

PRACTICAL WEEK – 2

```
1 • DROP TABLE IF EXISTS Loans;
2 • CREATE TABLE Loans (
3     loan_id INT PRIMARY KEY,
4     member_id INT,
5     book_id INT,
6     loan_date DATE,
7     FOREIGN KEY (member_id) REFERENCES members(member_id),
8     FOREIGN KEY (book_id) REFERENCES books(book_id)
9 );
```

```
INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
```

```
(1, 1, 4, '2025-01-05'),
(2, 2, 5, '2025-01-08'),
(3, 3, 6, '2025-01-10'),
(4, 1, 5, '2025-02-01'),
(5, 2, 4, '2025-02-05'),
(6, 3, 5, '2025-02-12'),
(7, 1, 6, '2025-03-01'),
(8, 2, 6, '2025-03-07'),
(9, 3, 4, '2025-03-15'),
(10, 1, 4, '2025-04-01');
```

Result Grid				
Filter Rows:				
	loan_id	member_id	book_id	loan_date
▶	1	1	4	2025-01-05
	2	2	5	2025-01-08
	3	3	6	2025-01-10
	4	1	5	2025-02-01
	5	2	4	2025-02-05
	6	3	5	2025-02-12
	7	1	6	2025-03-01
	8	2	6	2025-03-07
	9	3	4	2025-03-15
	10	1	4	2025-04-01
•	NULL	NULL	NULL	NULL

```

35 • SELECT
36     m.full_name AS Member_Name,
37     b.title AS Book_Title
38 FROM Loans l
39 INNER JOIN members m
40     ON l.member_id = m.member_id
41 INNER JOIN books b
42     ON l.book_id = b.book_id;

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```

43 • SELECT
44     published_year,
45     COUNT(book_id) AS Total_Books
46 FROM books
47 GROUP BY published_year
48 ORDER BY published_year;

```

- ✓ 15 12:23:29 INSERT INTO books (book_id, title, isbn, published_year) VALUES (7, 'Animal Farm', '9780451526342', 1945) 1 row(s) affected
- ✓ 16 12:23:29 INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date) VALUES (1, 7, 'Raj Kum...' 1 row(s) affected
- ✓ 17 12:23:29 COMMIT 0 row(s) affected

Result Grid				
Filter Rows:				
	donation_id	book_id	donor_name	donation_date
▶	1	7	Raj Kumar	2026-08-14
•	NULL	NULL	NULL	NULL

```

68 • CREATE INDEX idx_books_isbn ON books(isbn);
69
70 • SELECT *
71 FROM books
72 WHERE isbn = '9780451524935';

```

Result Grid				
Filter Rows:				
	book_id	title	isbn	published_year
▶	NULL	NULL	NULL	NULL
•	NULL	NULL	NULL	NULL